

Mwalimu wa Grade 10

Night-before teaching packs for a Grade 10 teacher preparing a strand she was never trained for, with no textbook and no internet in the room. Built online at night, used entirely offline in the classroom, holding no learner data.

What we are actually solving

A curriculum design document tells a teacher the **outcome** — "the learner should be able to explain X" — and never tells her X. With no textbook delivered, there is nowhere for her to go and learn X. So three problems are stacked inside one brief:

1. **She doesn't know the content.** The design states outcomes, not subject matter.
2. **She can't deliver it with what's in the room.** The design's suggested activities quietly assume a lab, a library, a projector, or internet research. She has chalk and fifty learners.
3. **She has to defend the plan.** Schemes of work get inspected. A plan she can't source is a liability.

The rule that shapes every decision we make today:

She cannot verify our output. Nobody in that room can. A farmer knows if the depot turned him away; a nurse knows a bad history when she reads one. A teacher who was never trained on this strand has no way to detect that we invented a plausible-sounding falsehood — and she will teach it to fifty children.

So **provenance matters more than polish.** Every line in a pack either traces to the design document with a page number, or is visibly labelled as generated. This is not a disclaimer we bolt on at the end. It is the product.

The artifact: one Night-Before Pack

Scoped to one sub-strand, one lesson. Eight sections, fixed order — and the order *is* the product, because understanding has to come before delivery.

SECTION	WHAT IT ANSWERS	SOURCE
1. What this actually is	The subject matter, pitched at a competent adult who has never seen the topic. Worked examples. The misconceptions <i>she</i> probably holds.	generated
2. Why it's here	Learning outcomes and the key inquiry question, quoted verbatim and cited to a page.	design
3. Tomorrow's 40 minutes	Minute-by-minute: hook, board work, activity, check, closure.	template
4. Board plan	What to draw and write, in order, in chalk, in under 90 seconds — and what to say while drawing it.	generated
5. The activity, with nothing	Fifty learners, no textbook, no lab, no printing, no power. Materials from the local market.	generated
6. They will ask you this	The questions learners actually ask, answered — plus a dignified script for admitting you don't know.	generated
7. How you'll know they got it	Three quick checks and a four-level rubric.	template

8. What to verify	Which lines came from the document and which we generated. Honesty rendered as a section.	—
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Architecture in one line

A **structured curriculum tree** feeds **six narrow generators**, whose output is a **self-contained pack** that lives on the device and prints to paper.

Three decisions follow from that, and they are the ones that will save or sink the day:

- **Parse the curriculum into structure — don't retrieve over PDF chunks.** The designs are already structured. A parsed tree gives exact, citable nodes; chunked retrieval gives fuzzy matches and citations we can't stand behind. The corpus is a few dozen sub-strands — it's a JSON file, not a vector database.
- **Six small generators, never one mega-prompt.** Each has its own contract, its own validation, its own risk profile. A partial failure still ships a usable pack; a monolith fails whole.
- **Offline is a file problem, not a runtime problem.** Once downloaded, a pack needs no server, no API, no model. Then offline isn't something we engineered — it's a property we get free.

Ownership map

Stay inside your own directories. Need something in someone else's? Ask them — don't reach in.

#	WORKSTREAM	OWNS
1	Curriculum & Data	data/curriculum/ · backend/db/ · backend/src/modules/curriculum/
2	Generation Engine	backend/src/modules/generation/
3	Pack API, Cache & Export	backend/src/modules/packs/
4	Prep Flow (frontend)	frontend/src/features/ · api/ · components/
5	Offline, Print & Share	frontend/src/offline/ · print/ · public/
—	Frozen	shared/ · docker/ · root configs — changes need a group decision

Member 1 — Curriculum & Data

You own the grounding source. Everything downstream cites you.

1. Hand-enter the corpus. One subject, one strand, three to six sub-strands. Outcomes, inquiry questions and suggested experiences **verbatim**, each with a real page number.

DONE WHEN: the JSON matches `shared/src/curriculum.ts` and every node has a page.

2. Idempotent seed loader. `npm run seed` loads `data/curriculum/*.json` into Postgres with `ON CONFLICT DO UPDATE`. You'll run it twenty times today.

DONE WHEN: running it twice in a row changes nothing and errors nowhere.

3. Curriculum read API. The four GET endpoints in `shared/src/api.ts`. Every object carries its `SourceRef` — a query that drops `doc_page` is a bug, not an omission.

DONE WHEN: you can walk subject → strand → sub-strand → detail with curl.

4. Tag `assumedResources`. For each suggested experience, list what it quietly assumes she has: internet, laboratory, textbook, projector, printer, field trip. This single field is what Member 2's activity translator runs on.

DONE WHEN: every experience has a list, and empty means "needs no translation".

5. Citation formatter. One helper that renders a `SourceRef` as *"KICD CSL Grade 10 Design, p. 24"*. Used by the UI and the PDF export so a citation never renders two different ways.

DONE WHEN: frontend and export both call it; neither formats a citation itself.

Member 2 — Generation Engine

You own the only code that talks to the model — and the section she can't check.

1. Briefing generator. Teaches the *teacher* the content. Effort `max` — this is the one section with no document to fall back on, so it carries the whole verification risk. Scope it hard to the outcome text; include the misconceptions she is likely to hold herself.

DONE WHEN: someone who doesn't know the topic can read it and teach the lesson.

2. Activity translator. Takes a real suggested experience plus its `assumedResources` and her actual resource level, and rewrites it for her room. Grounded by construction — the input is a real document line. Every material free or a few shillings, plus a no-materials fallback.

DONE WHEN: the learning intent is identical and nothing on the list needs a shop.

3. Board plan generator. One to four chalk panels: what to draw, what to say while drawing it, optional ASCII sketch. The blackboard is our render target — there is no projector.

DONE WHEN: you could stand at a board and follow it without rereading.

4. Questions and rubric generators. Three to eight questions learners really ask, answered, plus the don't-know script. Quick checks and a four-level rubric row per outcome.

DONE WHEN: both return schema-valid output with no missing rubric level.

5. Orchestrator with graceful degradation. `Promise.allSettled` across the generators; each rejection becomes a warning and a missing section, never a failed request. **Your code stamps provenance, never the model** — the model does not get to tell us how trustworthy it is.

DONE WHEN: you can break one generator on purpose and still get a usable pack.

Member 3 — Pack API, Cache & Export

You own the artifact leaving the building — and the demo's insurance policy.

1. POST /api/packs end to end. Validate, check cache, load sub-strand, generate, persist, return.

DONE WHEN: one curl produces a complete stored pack.

2. The cache. `sha256(sub-strand + context)`, unique index, DB hit before any API call. The second teacher to ask about a sub-strand pays nothing and waits nothing.

DONE WHEN: an identical repeat request returns instantly with zero API calls, and you can show that.

3. Self-contained HTML export. One file, inline CSS, zero external requests. This is what makes offline free and makes teacher-to-teacher sharing need no infrastructure at all.

DONE WHEN: the file opens correctly from the filesystem with wifi off.

4. PDF export. Server-side render of that same HTML. Two sides of A4 — there is no power at the front of her classroom, so paper is a real output format.

DONE WHEN: it prints legibly on physical A4 without cutting a section in half.

5. Demo pre-warm script. `npm run prewarm` generates and caches the exact demo path. Then a flaky venue network, a rate limit, or an expired key cannot touch the pitch.

DONE WHEN: the demo path loads from cache with the API key removed entirely.

Member 4 — Prep Flow (frontend)

You own everything she touches. Three taps from open to pack.

1. The picker. Subject → strand → sub-strand. Build it against `GET /api/packs/sample` from minute one — don't wait for Members 1 and 2.

DONE WHEN: three taps reach a sub-strand on a phone-width screen.

2. Context form. Class size, lesson minutes, language, resource level — all with defaults good enough to skip. Every extra field costs a tap, and she is doing this at 9pm.

DONE WHEN: she can go straight from picker to Build without typing anything.

3. Pack view. All eight sections in the fixed order, phone-shaped, skimmable in four minutes. If it can't be skimmed at 9pm it won't be read at all.

DONE WHEN: the sample pack renders completely, top to bottom, on a phone.

4. Provenance chips. Every section labelled *from the design / generated*, with the citation line where one exists. This is the visible form of the project's core rule — it is a feature, not a footnote.

DONE WHEN: no section renders without a provenance chip.

5. Honest progress. Generation takes 20–40 seconds. Name what's being built as it lands, section by section. Never a bare spinner, never a blank screen.

DONE WHEN: a first-time user never wonders whether it has hung.

Member 5 — Offline, Print & Share

You own the claim the whole pitch rests on. Your work is the demo's spine.

1. Installable PWA. Service worker precaches the app shell; installs to an Android home screen. No app store, no download, no 40MB APK on her data bundle.

DONE WHEN: it installs on a real phone and opens from the home screen.

2. IndexedDB pack store. Save on generate, read on open. **Local first, network as fallback** — offline is the expected state in her classroom, not the error case.

DONE WHEN: opening a saved pack makes no network request at all.

3. Airplane-mode proof. The `verifiedOffline` badge, set once a pack has been opened with no network. And the offline banner reads "your saved packs still work" — never an error.

DONE WHEN: wifi off, hard reload, pack still opens. Rehearse this one.

4. Print stylesheet. Whole pack on two sides of A4. Sections never split across a page break. Provenance survives to paper — a printed claim with no source is exactly the failure we exist to prevent.

DONE WHEN: it prints on real A4 and you'd teach from the printout.

5. Share. Web Share API to push the exported file to WhatsApp or Bluetooth, with a plain download fallback. Zero-infrastructure teacher-to-teacher distribution — sneakernet as a feature.

DONE WHEN: a pack moves from one phone to another with no server involved.

Contracts and merge protocol

- **shared/src/* is frozen.** It's the reason five people can work at once. Changing it needs a group decision, and everyone runs `npm run build:shared` after pulling.
- **Branch per member** — `m1-curriculum`, `m2-generation`, and so on. Touch only your own directories.
- **Merge continuously, not at the end.** Anyone still sitting on a six-hour-old branch at hour five is a merge disaster with a countdown on it.
- **main must always start.** If your merge breaks `docker compose up`, you stop and fix it before anything else.
- **Build against `GET /api/packs/sample`.** Nobody waits for anybody.

Shape of the day

hour	what happens
0–0.5	All five run <code>docker compose up</code> and see the sample pack render. Nobody moves on until it works for everyone.
0.5–2	M1 corpus · M2 briefing prompt against the fixture · M3 cache + endpoint · M4 pickers · M5 PWA install
2	First merge to main.
2–5	Core features. First real generated pack by hour four.
5	Second merge. Feature freeze on anything not in the demo path.
5–7	Export, print, offline verification, polish.
7	Hard freeze. Pre-warm the cache.
7–9	Rehearse the demo three times. Fix only what breaks the demo.

The demo, 90 seconds

1. "It's 8:47pm. Grade 10, period two tomorrow. She has never taught this strand, and there is no textbook."
2. Three taps: subject, strand, sub-strand.
3. The pack appears. Scroll the briefing. *"This teaches her, not the class. That's the part everyone else skips."*
4. Point at a provenance chip. *"This line is from the design, page 24. This one we generated — and we say so, because she has no way to check us."*
5. **Turn on airplane mode. Hard reload.** Still there.
6. Print preview. Two sides of A4, because there's no power at the front of the room.
7. *"No child logs in. There is no student table in our schema. Audit it."*

Four things we are deliberately not building

- **No learner-facing anything.** No child logs in, no names, no submitted work, no photographs of exercise books. It's structural, not a policy promise — and one "share it with students" feature would disqualify us.
- **No vector database or embeddings.** A few dozen sub-strands is a JSON file. Chunked retrieval would also break our citations, which are the product.
- **No PDF parser.** Automated table extraction on these designs will eat four hours and produce mush. Manual entry of six sub-strands takes under an hour and is flawless. In a one-day build, typing is the correct engineering decision.
- **No on-device model.** "No internet in the room" is not "no internet ever". She preps at home on data; the *artifact* is what goes offline.

Elimu · Mwalimu wa Grade 10 — team brief. Repo `README.md` has setup; `data/curriculum/README.md` has the corpus format. The sample pack fixture in `shared/src/fixtures/` contains **invented page numbers** — never demo with citations that came from it.